MILITARY SPECIFICATION

BAR, ALINING, BRIDGE TRUSS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bridge truss alining bar.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MILITARY

MIL-T-704 - Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED. STD. No. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/ASTM A 120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses.

ANSI/ASTM A 576 - Steel Bars, Carbon, Hot-Rolled, Special Quality.

ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The truss alining bar shall be as shown in figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein (see 6.4 and 6.5). Dissimilar metals as defined in MIL-T-704 shall not be used in intimate contact with each other unless the metals are protected against galvanic corrosion.

3.2.1 Steel pipe. Steel pipe shall conform to ANSI/ASTM A 120, standard weight, black finish, unthreaded ends, except that the hydrostatic pressure test is not required.

3.2.2 Steel rod. Steel rod shall conform to ANSI/ASTM A 576, UNS G10180, round form.

3.3 Treatment and painting. The alining bar shall be cleaned, treated and painted in accordance with MIL-T-704, type B, except that the color shall conform to FED. STD. No. 595, color No. 34087.

3.4 Workmanship. The truss alining bar shall have a smooth surface, free of burrs, rough edges, or foreign matter which may affect its use.

3.4.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the truss alining bar to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the truss alining bar shall be at the option of the contractor.

3.4.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.
ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME code

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3.4.3 Weld practices.

a. Preheat of materials being welded, and maximum interpass temperature during welding, shall be in accordance with the contractor's recommendations.

b. Work shall be positioned for flat welding whenever practicable.

c. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum.

3.4.4 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Material inspection. The contractor is responsible for insuring that materials used are manufactured and examined in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level III.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 4.0 percent defective for major defects and 10 percent defective for minor defects.

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4.4 Inspection procedure.

4.4.1 Examination. The truss alining bars shall be examined as specified herein for the following defects:

Major

101. Truss alining bar not as shown in figure 1.
102. Material not as specified.
103. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

104. Containers not as specified for level A or B.
105. Quantities packed together exceed the limitations of the box for level A or B.
106. Bars not secured within the box in a manner to prevent movement for level A or B.
107. Packing not in accordance with the referenced document for commercial.
108. Marking missing, illegible, incomplete, or incorrect for level A, B, or commerical.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

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5.1.1 Level A. The bars shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style A or B; or PPP-B-621, class 2, style 2; in quantities not to exceed the weight limitation of the box. The bars shall be blocked and braced within the box in a manner to prevent movement. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified, strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.1.2 Level B. The bars shall be packed together as specified in 5.1.1 except boxes shall be domestic type or class, as applicable. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Commercial. The bars shall be packed in accordance with MIL-STD-1188.

5.2 Marking.

5.2.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The alining bar is intended for alining truss members of fixed, single-lane panel bridges during erection.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When welding is to be done by means other than the contractor's standard practices (see 3.4.1).

c. Degree of packing required (see 5.1).

d. When other than finish B strapping is required (see 5.1.1).

6.3 Packaging. Preservation is unnecessary. For purposes of preservation/packing level marking, preservation shall be designated level A.

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6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.5 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bids, requests for proposal, and contracts the following statement:

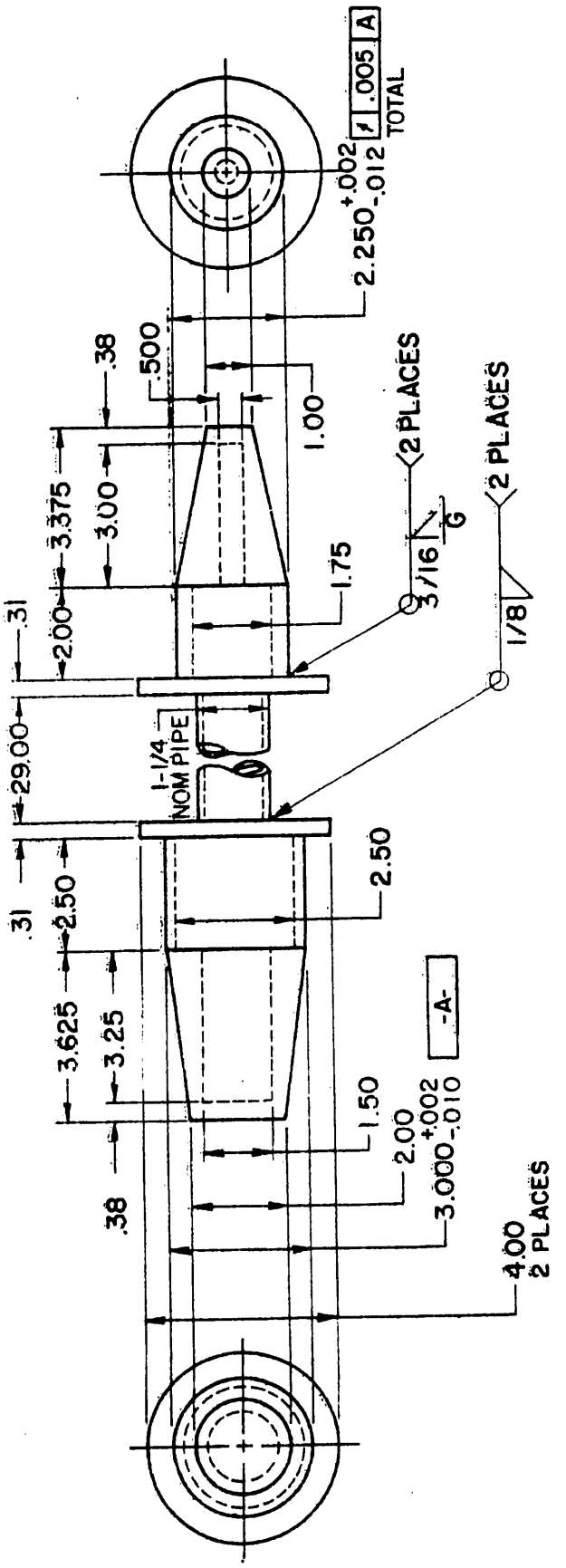
"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materiels, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed." (see 3.2).

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NOTE:
1. ALL DIMENSIONS ARE IN INCHES.
2. TOLERANCES UNLESS OTHERWISE SHOWN: .XX ± .06 .XXX ± .030

X-2670A

FIGURE 1. Bridge truss aligning bar.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-52785A(ME) Bar, Alining, Bridge Truss

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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